Reason. Plan. Act. Physical AI Workshop

Lecture 1a: Introduction

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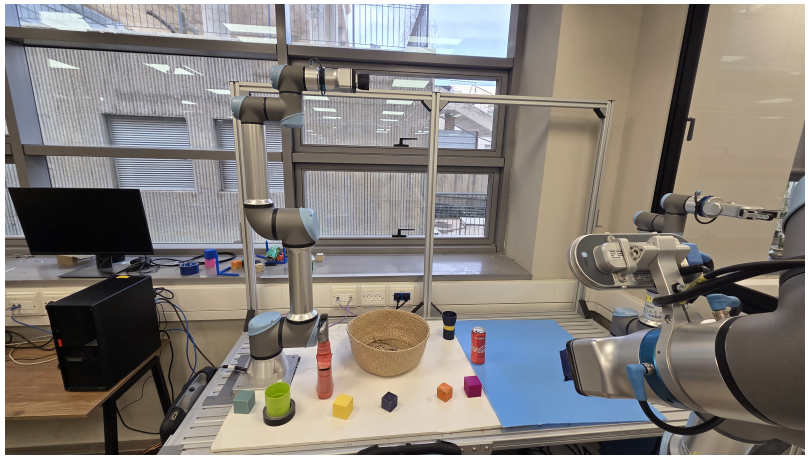
The Taub Faculty of Computer Science
Technion - Israel Institute of Technology

Physical AI

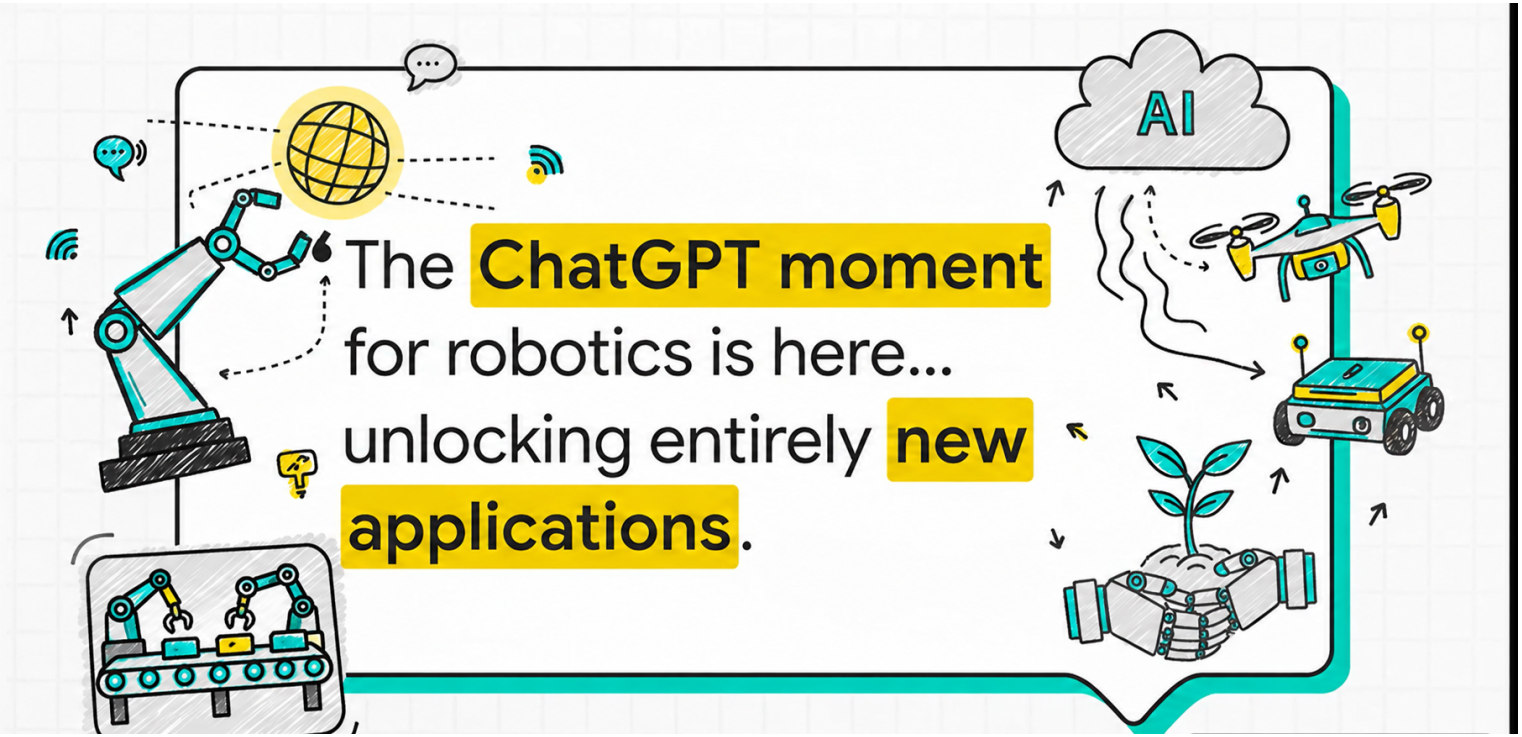
https://youtu.be/cRZNwgvcWUg?si=kIfNi9zTUdHsXraJ&t=787

Physical AI



Objectives

Physical AI


AI
The ChatGPT moment
for robotics is here...
unlocking entirely new
applications.

Physical artificial intelligence

4 languages

Article Talk Read Edit View history

From Wikipedia, the free encyclopedia

Not to be confused with Physical Intelligence Inc.

Figure AI robot package handling

Physical artificial intelligence or **physical AI** refers to artificial intelligence (AI) systems that perceive, reason about and act within the physical world, specially at the human scale. These systems generally combine AI models with sensors, control systems, actuators and machines such as robots or autonomous vehicles. Physical AI overlaps with embodied artificial intelligence, robotics and autonomous systems, but it emphasizes the complete process of perceiving an environment, motion planning an action and physically executing the task to perform work. This differs from digital AI or generative AI (GenAI), which primarily stays in the information or digital realm.[1][2][3]

The term became increasingly prominent during the AI boom in the 2020s as AI development expanded from primarily digital applications toward humanoid robots, self-driving vehicles, smart factories and other autonomous machines.[4] Its boundaries are not standardized, and it is often treated as a continuation of earlier research in robotics and embodied intelligence rather than an entirely separate field.[5]

No universally accepted formal definition. Many different interpretations.

Objectives

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Example

Autonomy in Robotics

Physical AI

Task and Motion Planning

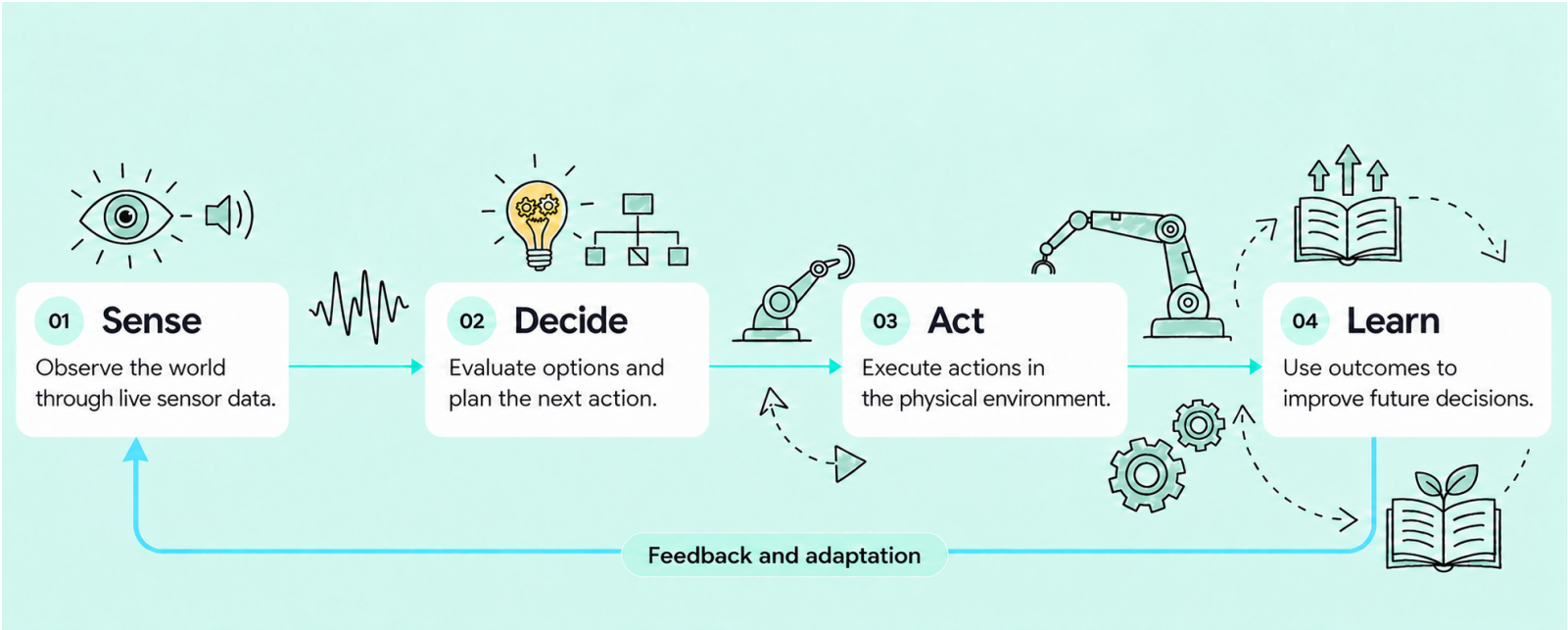
Physical AI

Unlike traditional abstract AI, Physical AI must account for the properties and constraints of the physical environment.

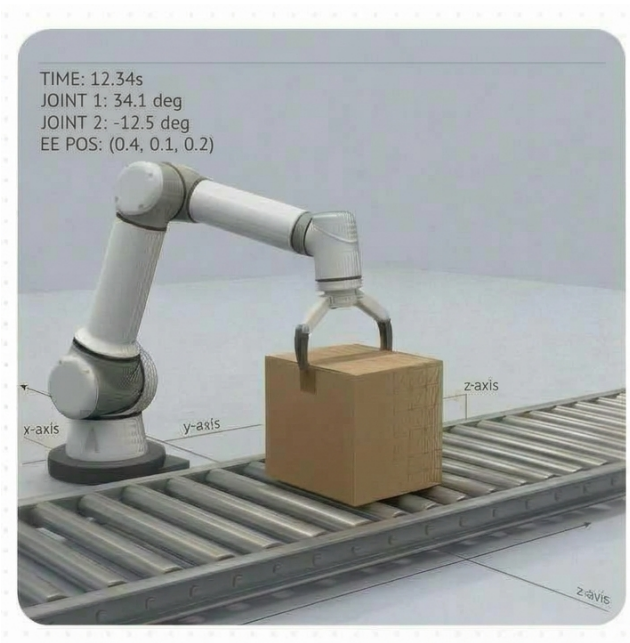
Currently a ‘hot’ topic, but underlying research builds on decades of work in robotics, embodied intelligence, planning, control, reinforcement learning, and autonomous systems.

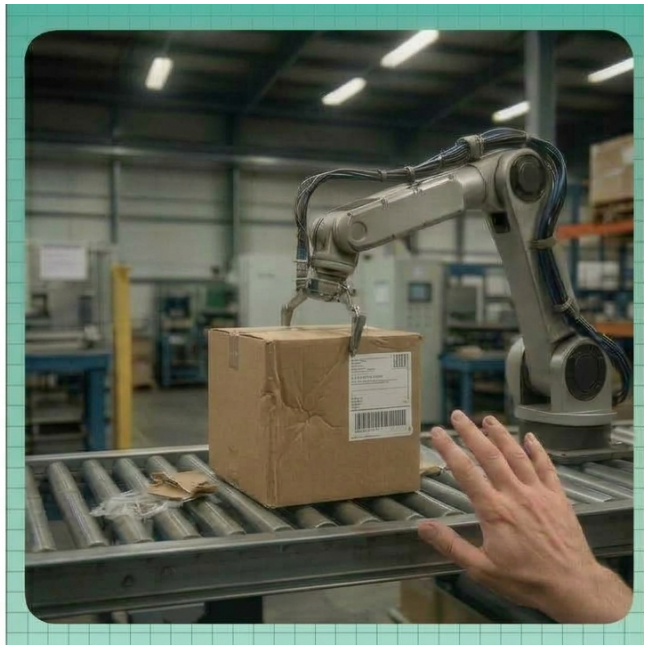
The Continuous Closed Loop

Continuous interaction between intelligence and the physical world.



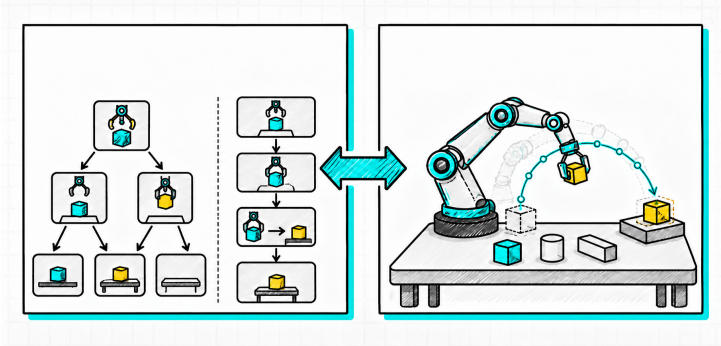
Sim-to-Real Gap





Our Objectives

Provide a practical understanding of how Physical AI systems combine **discrete task decisions** with **continuous** geometry, collision **constraints**, and **feasible** robot motion.



Account for **uncertainty and imperfect** sensing, changing environments, and execution **failures**.

Implement and evaluate core algorithms in simulation and on physical robotic platforms.

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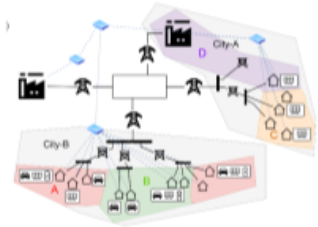
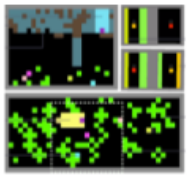
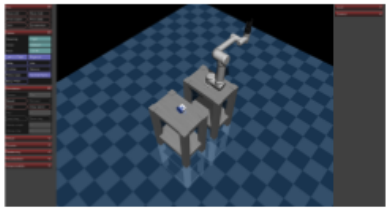
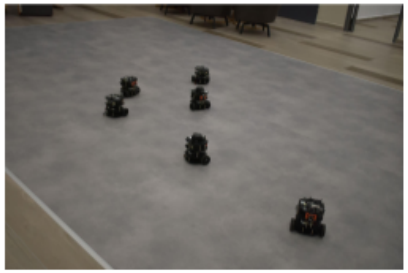






Collaborative AI and Robotics



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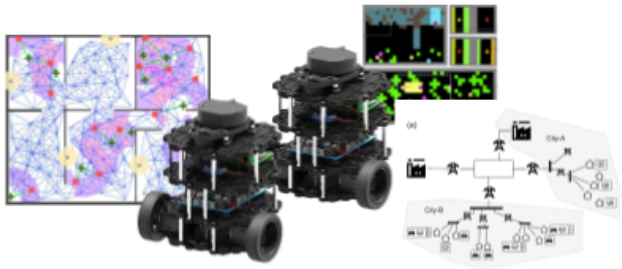
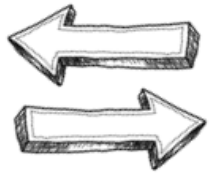
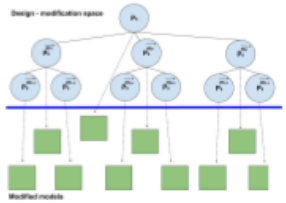
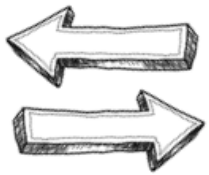




Provide theoretical foundations for designing AI systems that are capable of effective partnership in sustainable and efficient collaborations of automated agents as well as of automated agents and people.

experience (data)  knowledge

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Team: Sarah Keren

Head of the Collaborative AI and Robotics (CLAIR) lab

Developing novel approaches to identify AI agents' inherent capabilities and limitations and modify their environment to maximize performance and promote effective multi-agent collaboration.

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Objectives
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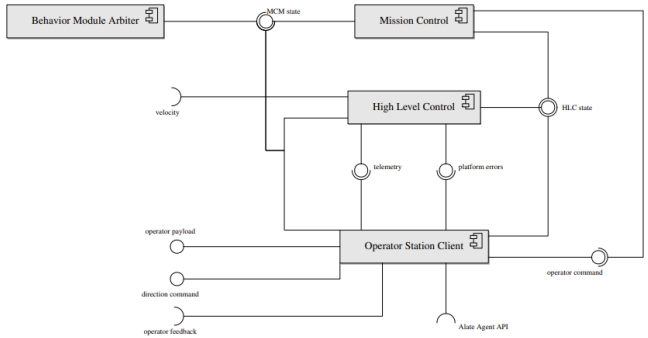
Example
Autonomy in Robotics
Physical AI
Task and Motion Planning

Team: David Dovrat



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Team: Guy Azran

Figure 1: Grocery task from Example 1.

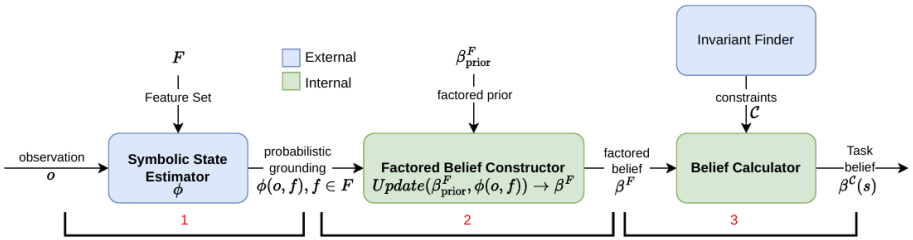
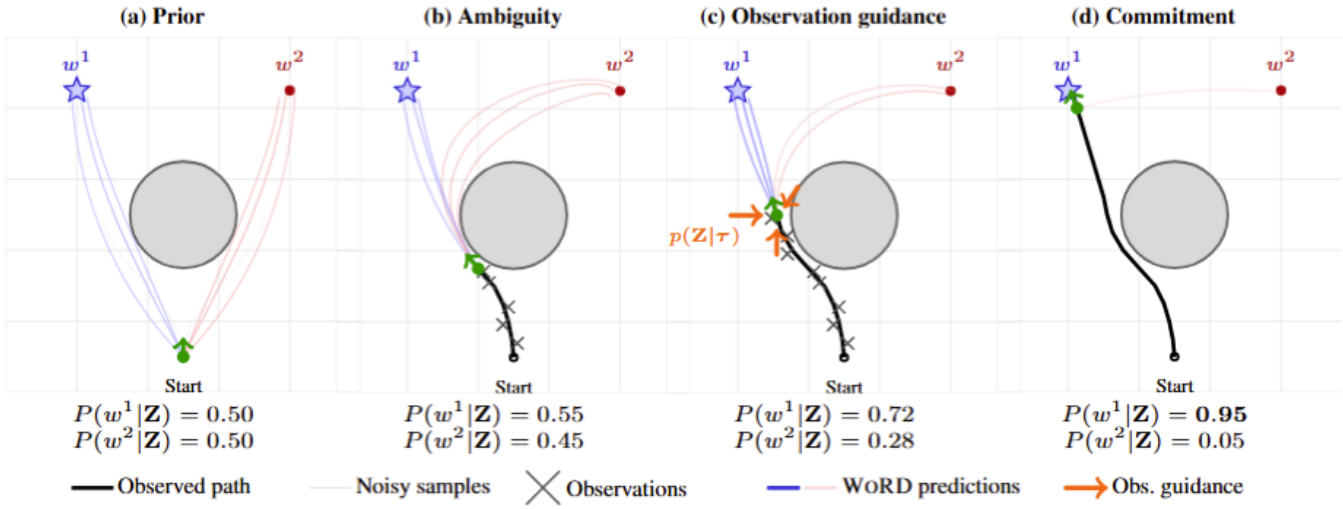


Figure 2: The three-layered belief pipeline. (1) An observation is processed by the SSE to obtain per-feature probabilities, (2) which are updated with an optional prior to produce the factored belief. (3) The factored belief and a set of task-specific constraints are then used to compute the belief.

Team: Almog Anschel



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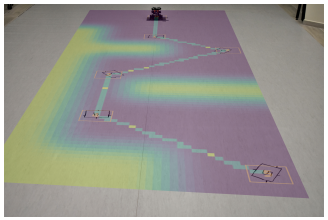
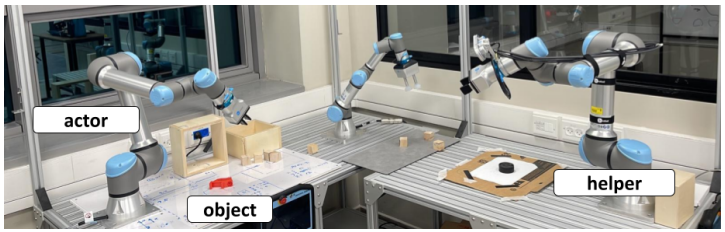
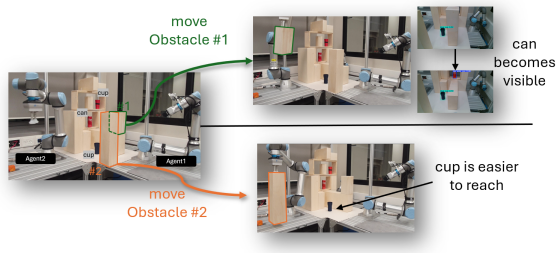
Work done at the CLAIR lab







Example

Example

”Pick up an apple”

Example

”Pick up an apple”

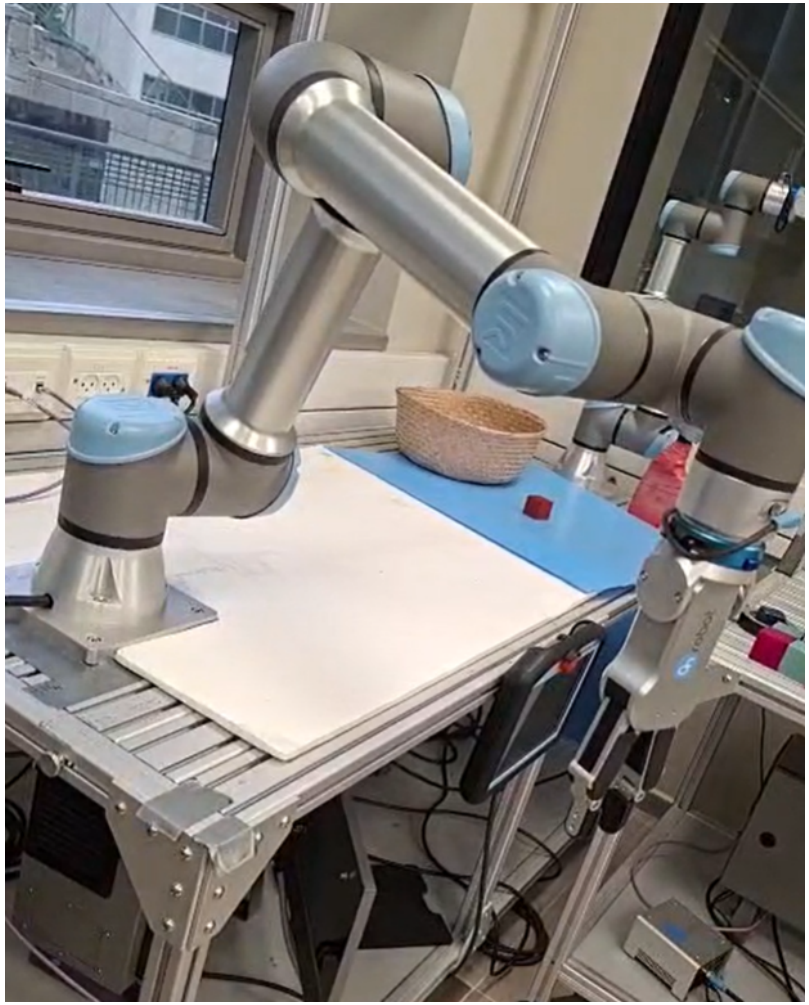
”I am hungry”

Example

Bluey and Bingo

https://youtu.be/n4rh2jD8OkY

CLAIR Demo



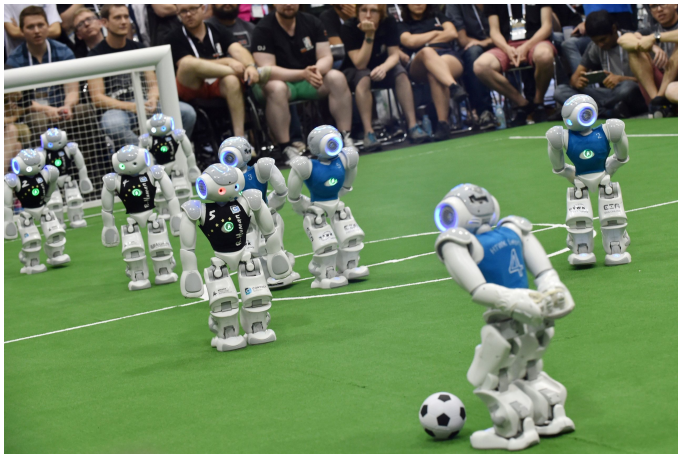
Autonomy in Robotics

Reactive Agents

Fixed policy based on sensor input

No internal model of the world

Fast but limited adaptability



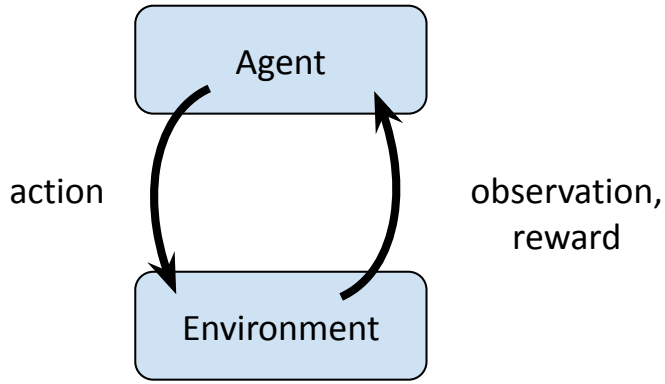
Rational Agents

Use planning and reasoning

Represent world knowledge

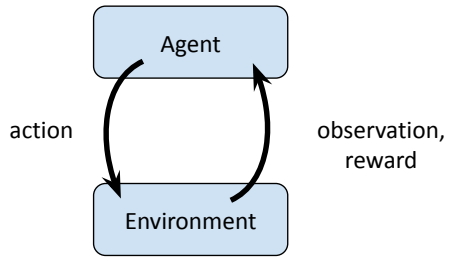
Flexible but computationally costly

Autonomy



Autonomy

- Action (Actuation)
  - Locomotion
  - Manipulation
- Perception (Sensors)
  - Internal
  - External
- Cognition (Control)
  - From reactive to proactive
  - From finite state machines to cognitive robotics



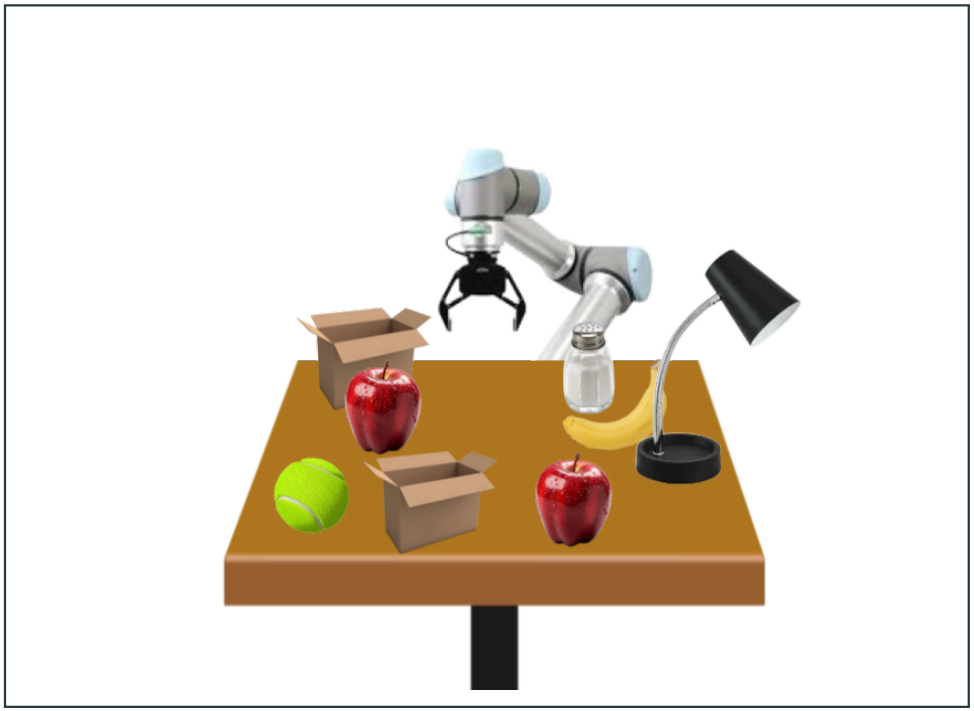
Approaches to AI Decision Making

We are seeking a **policy**, a (potentially probabilistic) mapping from the state space to actions.

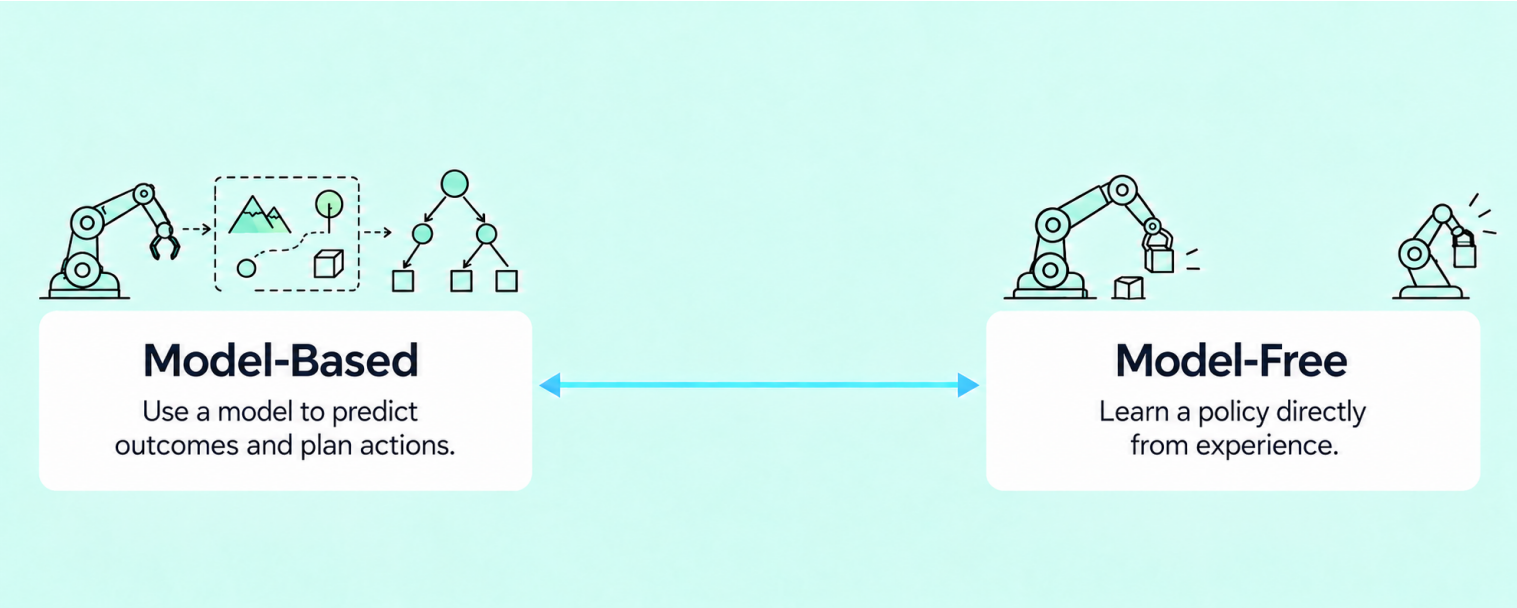
Basic approaches:

- **Programming/reactive** - fixed decision making
- **Planning** - using a domain-independent descriptive language
- **Machine Learning** - extracting a policy from data and experience
- **Reinforcement Learning** - learning by interacting with the environment

Best approach for our setting?



Model-free vs. Model-based



Model-free vs. Model-based

Model-based

- Has (or learns) a **model** of the world: $T(s' \mid s, a)$, $R(s, a)$
- Chooses actions by **planning** with the model (search, value iteration, MPC, MCTS)
- \+ Sample efficient, can reason about new goals
- – Model errors lead to bad plans (*model bias*)

Model-free

- No explicit model of the dynamics
- Learns a **policy** $\pi(a \mid s)$ or **value** $Q(s, a)$ directly from experience (Q-learning, policy gradient, PPO)
- \+ Simple, no modeling effort, strong asymptotic performance
- – Needs a lot of interaction data, hard to transfer to new tasks

Hybrids combine both: learn a model, then use for planning or training

Uncertainty

Uncertainty

Sources of uncertainty:

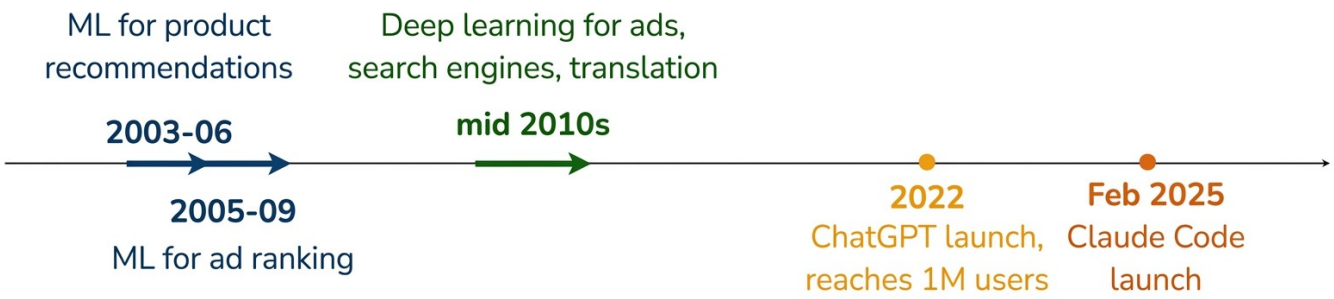
- **Perception** – noisy, partial sensor readings (*partial observability*)
- **Action** – actuators slip, grasps fail (*stochastic outcomes*)
- **Model** – dynamics and environment only approximately known
- **Other agents** – humans and other robots behave unpredictably

Uncertainty

Uncertainty

Physical AI

Timeline: Learning



Timeline: Robotics

- **Before ~2020**: Classical robotics: localization, mapping, model-based control, motion planning, task planning.
- **2021–2023**:Language-conditioned manipulation, large robot datasets, RT-1/RT-2, diffusion policies.
- **2024**: OpenVLA and larger cross-robot learning ecosystems accelerate open VLA research.
- **2025**: Gemini Robotics, on-device variants, $\pi 0$-style policies, and new embodied reasoning systems.
- **2026**: Stronger planning and spatial reasoning; expanding evaluation, data pipelines and commercialization.

Four technical paradigms

1. Classical / modular

Explicit state estimation, symbolic task planning, motion optimization and feedback control.

2. Imitation / offline policies

Behavior cloning, diffusion policies and action-chunking trained on demonstrations.

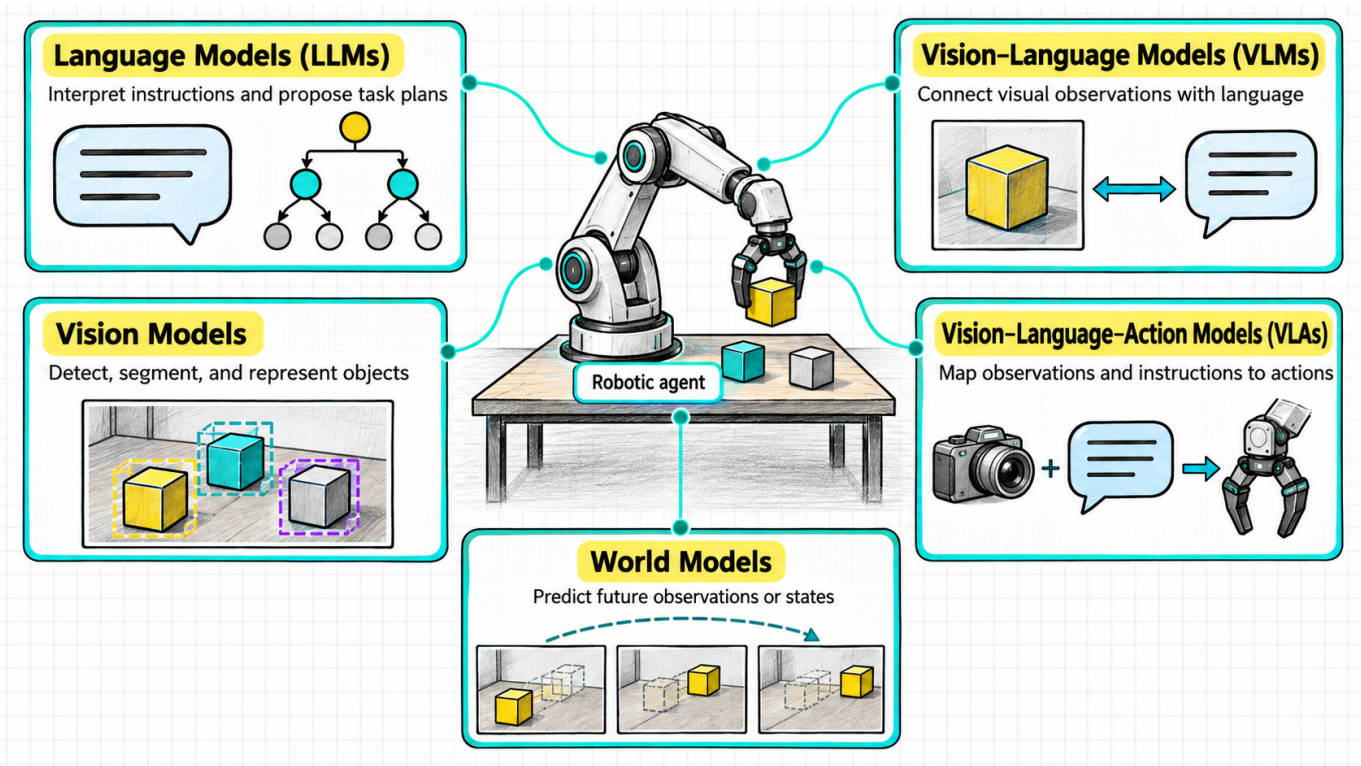
3. Foundation-model VLA

A multimodal model maps observations + language goals to actions or action tokens.

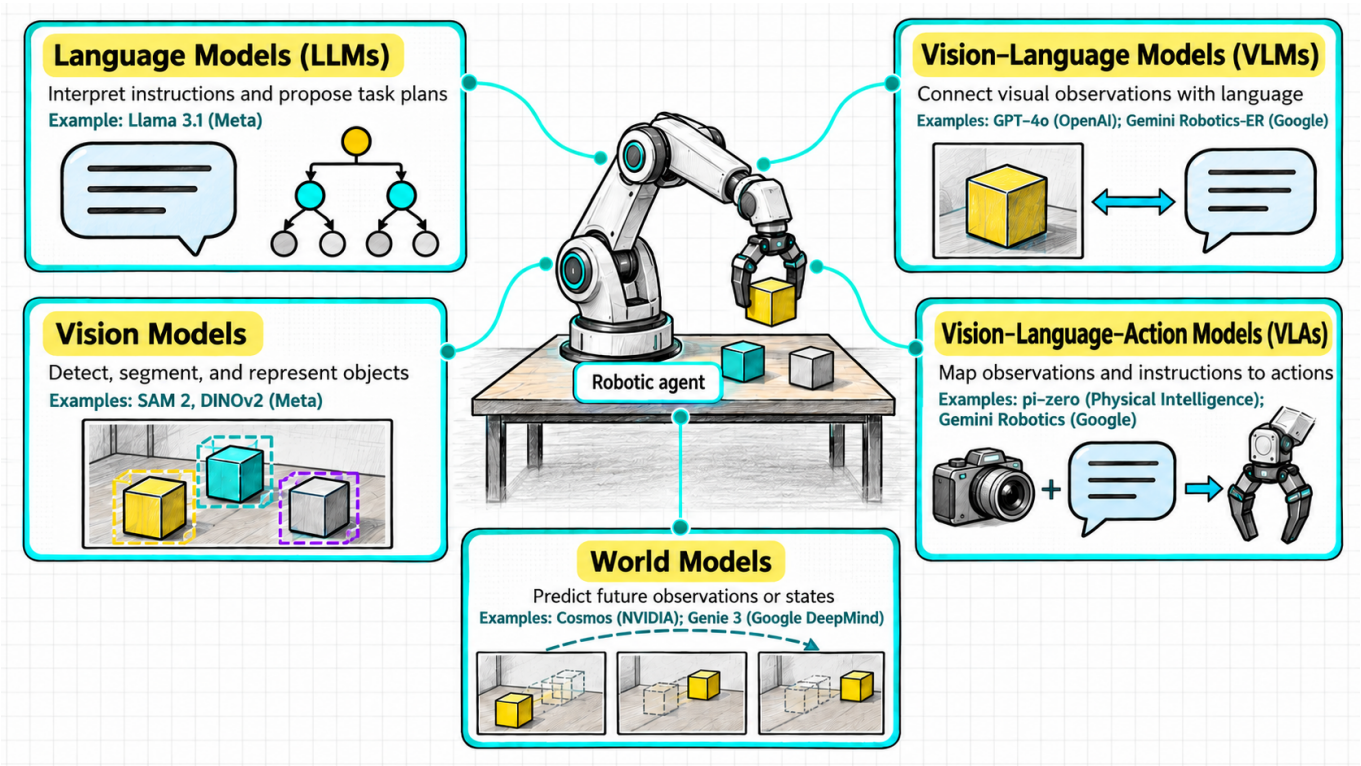
4. Hybrid agentic systems

Embodied reasoning orchestrates skills, tools, symbolic constraints and reactive policies.

Foundation Models for Robotics



Foundation Models for Robotics



Robotics Solved?

Robotics Solved?

Still unresolved

Reliable multi-hour autonomy, guarantees under shift, dexterity in unstructured homes.

Persistent bottlenecks

Long-horizon reliability, uncertainty, manipulation, safety and evaluation remain unresolved.

Data engines

Teleoperation, large-scale demonstrations, simulation and synthetic augmentation now shape progress.

Why physical intelligence is harder than text generation

- **Partial observability** Occlusion, sensor noise, ambiguous objects and hidden contact states.
- **Distribution shift** Rooms, illumination, tools, humans and hardware are never exactly repeated.
- **Continuous dynamics** Actions have geometry, torque, friction, latency and stability constraints.
- **Costly data** Physical trials require equipment, operators, resets and risk management.
- **Long-horizon coupling** An early bad grasp can invalidate a sequence of downstream actions.
- **Safety** A mistake may damage property or injure someone; monitoring matters.

Vision–Language–Action

Foundation models as planners: Vision-language-action (VLA) policies connect semantic instructions and perception to robot actions.

maps: (image, instruction) $\rightarrow$ **motor actions**

Inputs: Camera / depth / proprioception Natural-language task Optional robot history

VLA Policy: $\pi(a_{t:t+H} \mid o_{\leq t}, g)$

Outputs:

End-effector poses Joint commands Discrete skill selections

Challenge:

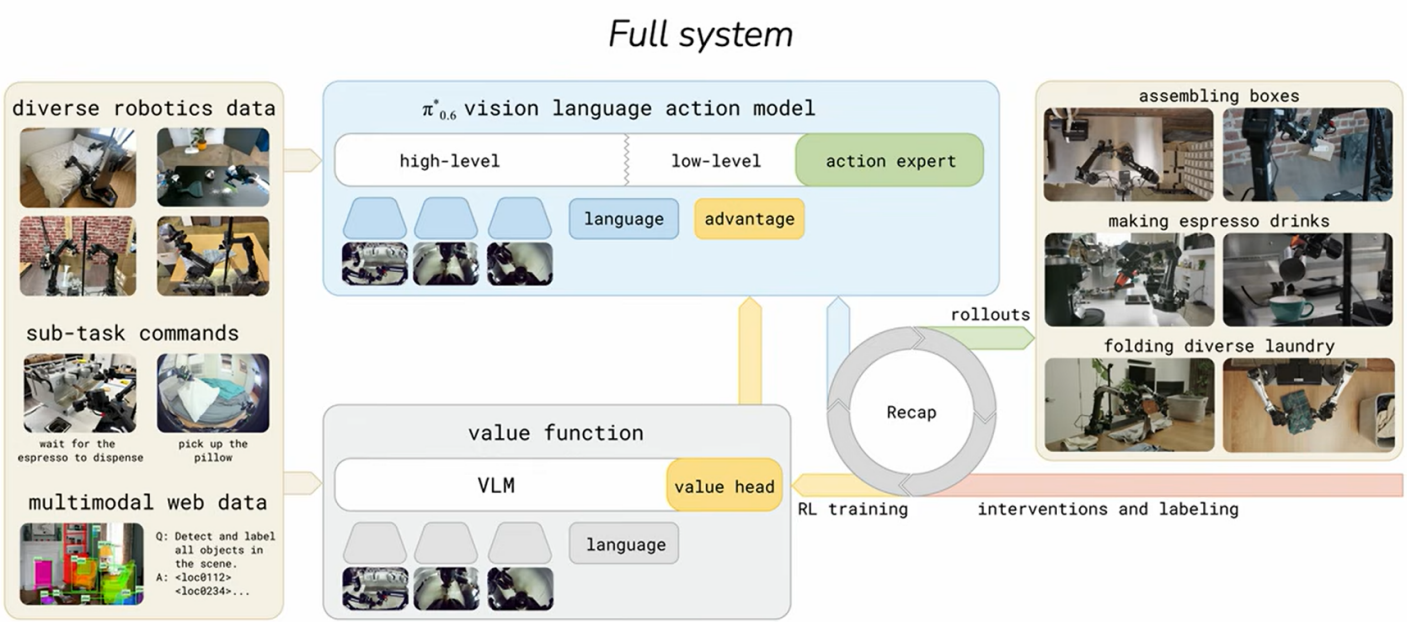
language-level intent must translate into continuous, contact-rich, feasible motion.

Foundation Models in Robotics

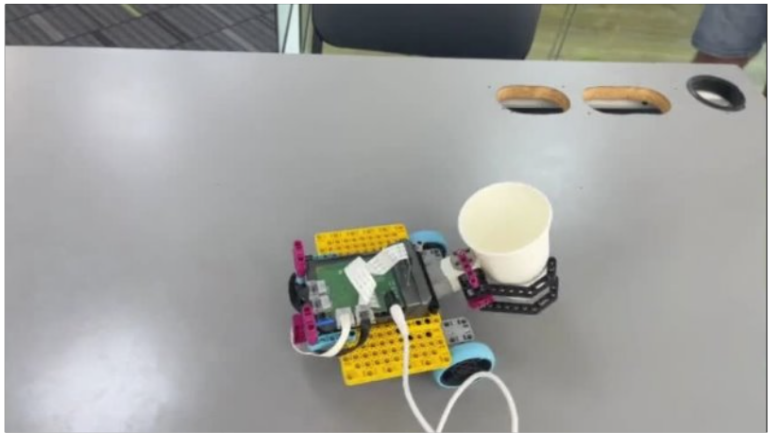
- Supervised learning on large demonstration datasets: $\max_\theta \ \mathbb{E}\big[\log \pi_\theta(a \mid o, \ell)\big]$
- \+ End-to-end, no hand-built model or skill library
- – Compounding errors off the data distribution; no explicit reasoning about goals or constraints

Trend: combine both (a planner reasons about *what* to do, a learned policy decides *how*)

VLA Structure



Teaching Neil new Skills



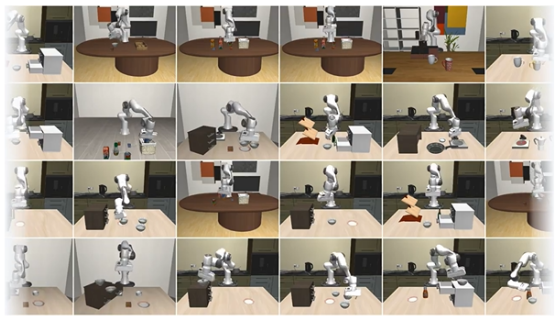
Acquiring new skills for mobile-manipulators

Why do we still to understand robotics ?

Why do we still to understand robotics ?

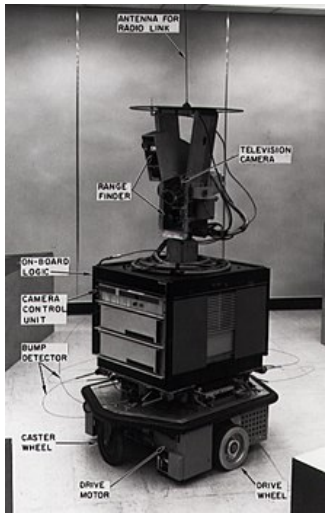
What are included

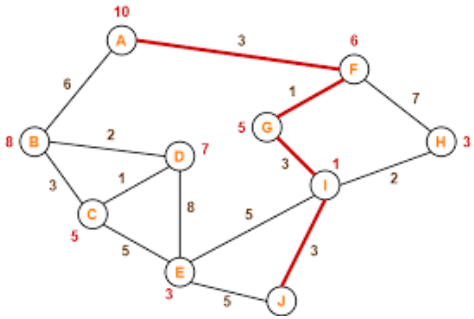
✓ RGB images from workspace and wrist cameras
✓ Proprioception
✓ Language task specifications
✓ PDDL scene descriptions



Task and Motion Planning

’go from A to B’?


ANTENNA FOR RADIO LINK
TELEVISION CAMERA
RANGE FINDER
ON-BOARD LOGIC
CAMERA CONTROL UNIT
BUMP DETECTOR
CASTER WHEEL
DRIVE MOTOR
DRIVE WHEEL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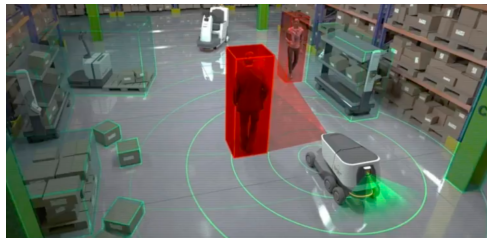
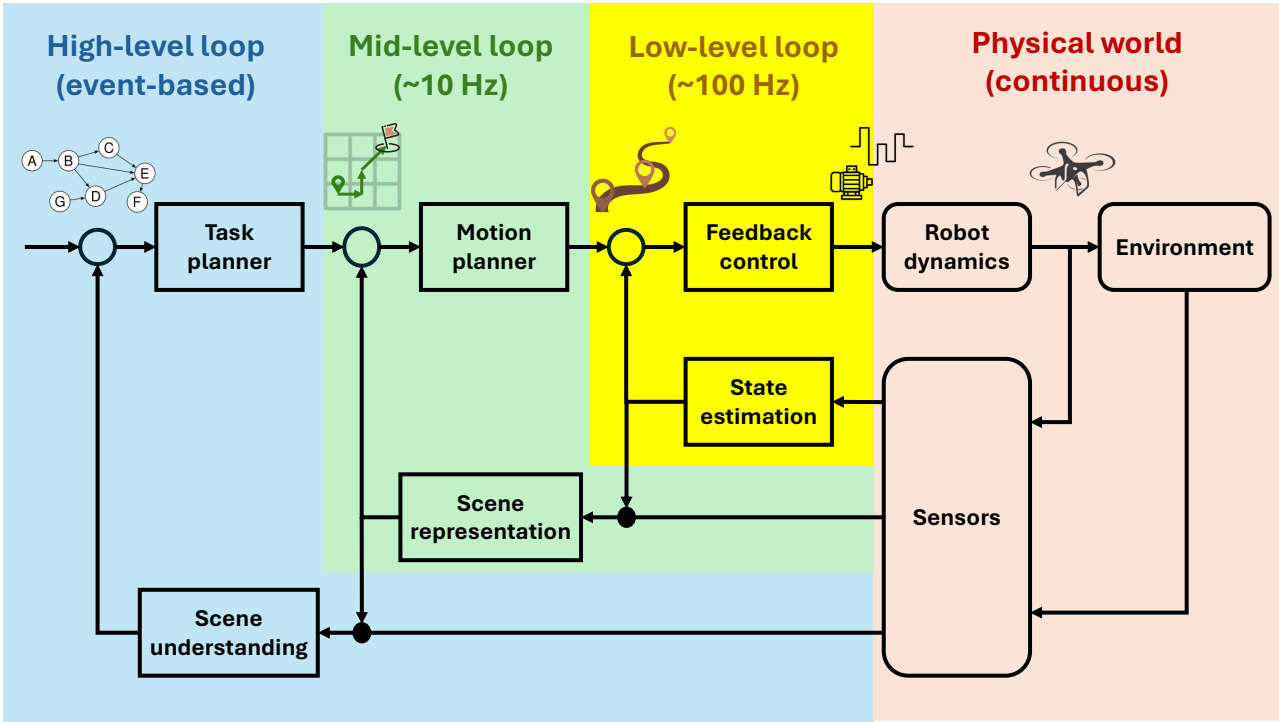


’Pick up an apple’

’I am hungry’

What about: ”I am hungry” ?

Layered Decision-Making



Perception and Scene Understanding

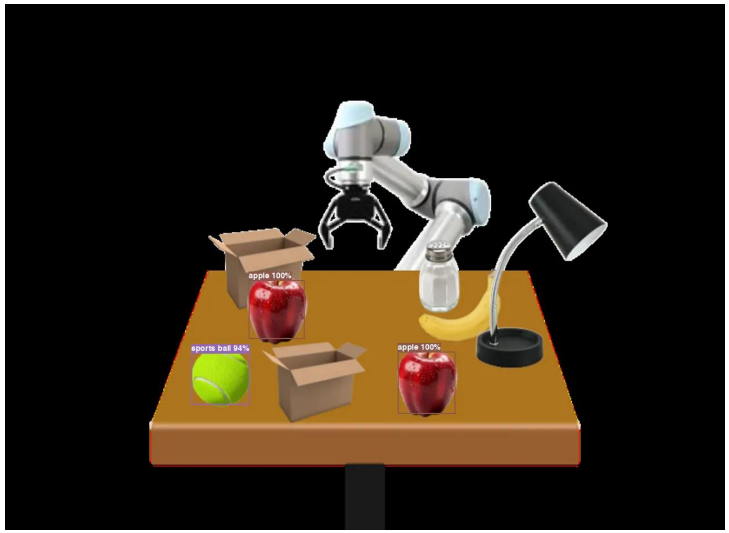
Sense and interpret the environment to locate an apple and estimate its properties.

Sensor Data Acquisition: for example, RGB-D camera (color + depth), LiDAR, stereo camera, tactile and proximity sensors...

`https://huggingface.co/spaces/nielsr/dpt-depth-estimation`

Perception and Scene Understanding

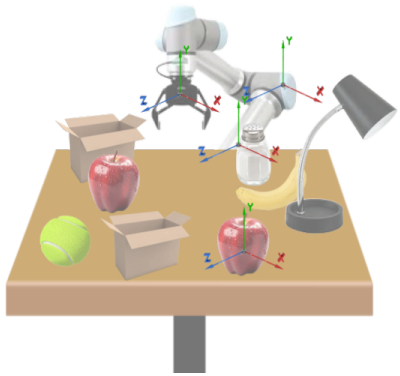
Object Detection and Segmentation: detect and classify objects in the image



YOLO v3: `https://v-iashin.github.io/detector`

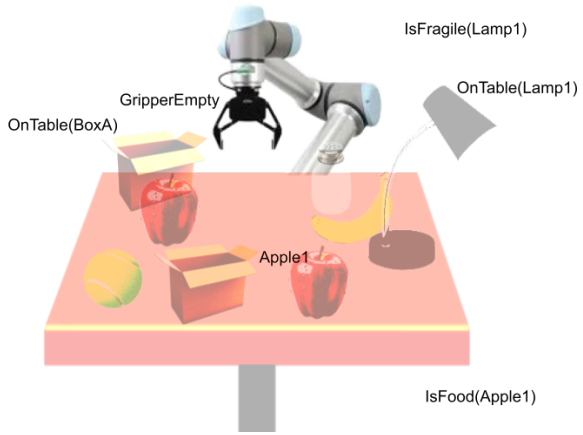
Perception and Scene Understanding

- **3D Localization:** Using the depth map, the robot computes the apple's 3D position in its camera coordinate frame.
- This point is transformed into world coordinates using the robot's calibration data.
- Uncertainty of estimation influences planning (e.g., grasp robustness).



Task Interpretation - Task Planning

Translate the scene into a symbolic representation



The representation (e.g., using PDDL or STRIPS domain) is designed to be used by the high-level symbolic planner that finds a sequence of actions or policy to achieve the goal.

Task Planning

- A planner translate the goal condition (“pick up an apple”) into a concrete sequence of actions.
- Possible representation:

Goal: (holding(apple) = true)

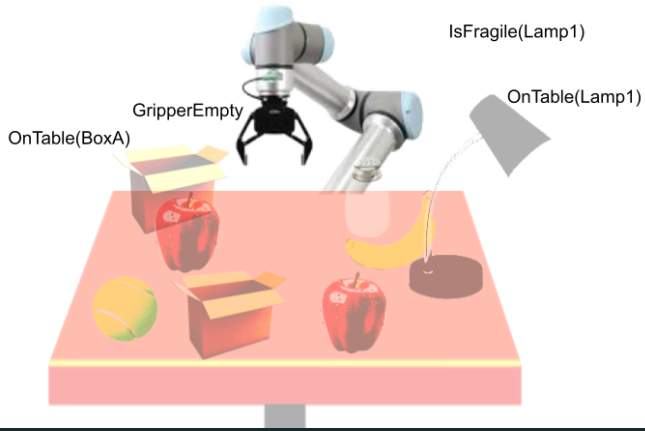
State Features: apple_on(table), gripper_empty ...

Actions: move_to(apple), grasp(apple)...

Is this a sufficient representation for our setting ? What if we have more than one way to achieve the goal (here, more than one apple)?

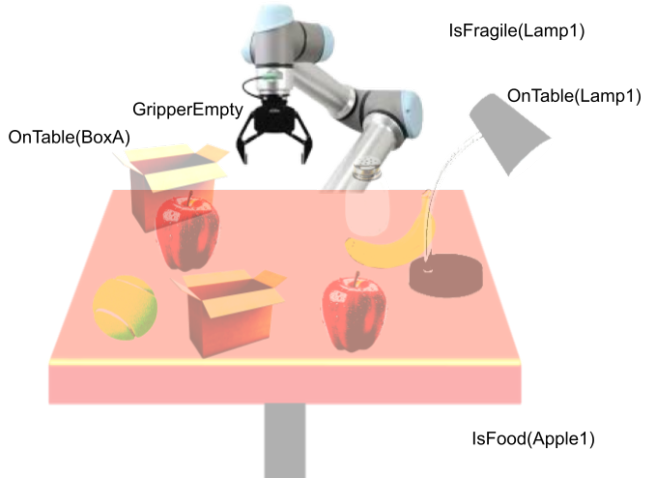
Task Planning

Let $\mathcal{R}$ be a robot operating in a discrete environment described by a set of symbolic **states** $\mathcal{S}$ and **actions** $\mathcal{A}$. Given an initial state $s_0 \in \mathcal{S}$ and a **goal condition** $G \subseteq \mathcal{S}$, determine whether there exists a sequence of actions $\pi = \langle a_1, a_2, \ldots, a_n \rangle$, such that executing π from s_0 leads to a state $s_n \in G$.



Task Planning

- **Input:** Goal condition: 'pick up an apple'
- **Output:** A sequence of (high-level) actions that achieve the goal
- An option: use propositional STRIPS to define the planning task as $P = \langle \mathcal{F}, I, \mathcal{A}, G, \mathcal{C} \rangle$ and use A* to solve the problem

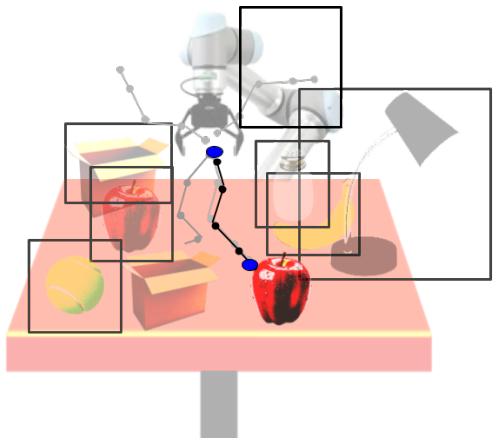


Task Interpretation - Motion Planning

A motion planner is used to compute a collision-free trajectory for the robot's arm to reach the apple.

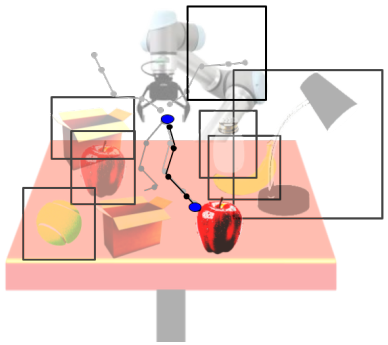
The planner ensures:

- desired position is reached (the gripper's approach vector aligns with the apple)
- no collision occurs with the table or nearby objects.
- kinematic feasibility (joint limits, reachability, etc.) is satisfied.



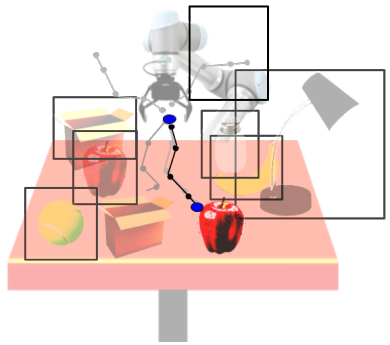
Motion Planning

Let $\mathcal{R}$ be a robot system with d **degrees of freedom**, moving in a known environment with obstacles. Given start and target **configurations** q_0 and q_g for $\mathcal{R}$, decide whether there is a collision-free, continuous path $\tau : [0, 1] \rightarrow \mathcal{C}_{free}$ such that $\tau(0) = q_0$ and $\tau(1) = q_g$ and if so, plan such a motion.



Motion Planning

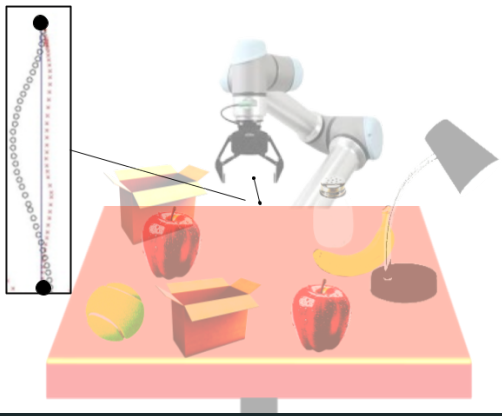
- **Input:** Initial and goal configurations and map of (obstacles in the) environment.
- **Output:** A collision-free path from the initial to the goal configuration.
- An option: map the environment into free and occupied space and use A* over the free space.



Control and Execution

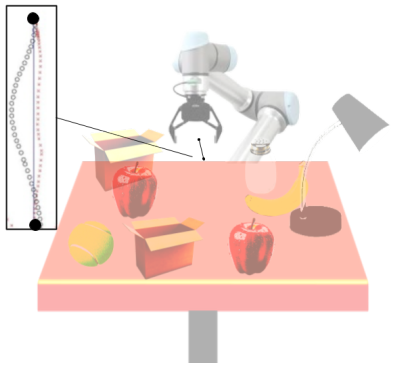
”Move the arm, close the gripper, and lift the apple”

- **Trajectory Execution** - low-level controller (e.g., PID, MPC) uses feedback from joint encoders and force sensors to follow the planned trajectory.
- **Force control loop** - tactile sensors measure contact pressure so that the controller can regulate the grasp force.
- **Vision-based feedback** - (visual servoing) refines the alignment if necessary; the apple disappears from the table view and appears in gripper view.



Motion Control

Given a desired trajectory $\tau(t)$ for a robotic system $\mathcal{R}$ with d **degrees of freedom**, the goal of motion control is to compute and apply control inputs $\mathbf{u}(t)$ (e.g., joint torques, velocities, or forces) such that the system's actual motion $q(t)$ follows $\tau(t)$ as closely as possible.



Motion Control

- **Input:** Desired trajectory
- **Output:** A control procedure that minimizes the deviation between the executed and desired trajectory.

System dynamics are captured by $\dot{x}(t) = f(x(t), u(t), t) + w(t)$ where

- $x(t) \in \mathbb{R}^n$ - state vector
- $u(t) \in \mathbb{R}^m$ - control input
- $f : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R} \to \mathbb{R}^n$ - system dynamics
- $w(t)$ - noise or disturbance

The challenge is to translate actions into low-level commands (and monitor their execution).

From Instructions to Torques

'pick up an apple'

- `pick_up_apple`
- `move_to_grasp_config`
- `follow_trajectory`

From Instructions to Torques

'pick up an apple'

- `pick_up_apple`
  - **State Estimation:** locate an apple in the scene
  - **Decision Making:** decide which apple to try and pick up
- `move_to_grasp_config`
  - **State Estimation:**
    offline: mark all the obstacles in the environment (including the robot) in the configuration space.
    online: detect robot position (in dynamic environments, also the position of obstacles)
  - **Decision Making:** decide how to reach the configuration
- `follow_trajectory`
  - **State Estimation:** monitor robot location w.r.t. reference trajectory
  - **Decision Making:** corrective actions

Uncertainty Management and Error Handling

- If the apple is occluded, the robot performs active exploration (moves head/camera).
- If the robot's gripper misses the apple, it may trigger a failure recovery behavior (e.g., retry with a new pose estimate).
- If grasp fails (slip detected or object dropped), the system may replan or retry.

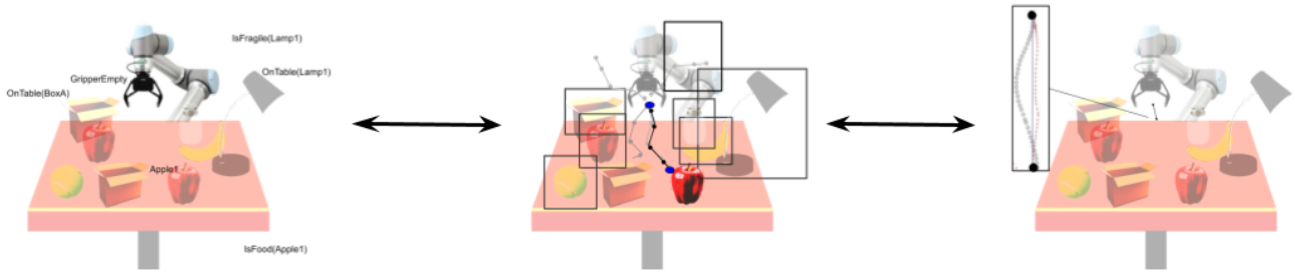
Learning and Adaptation

The objective is to improve performance over time.

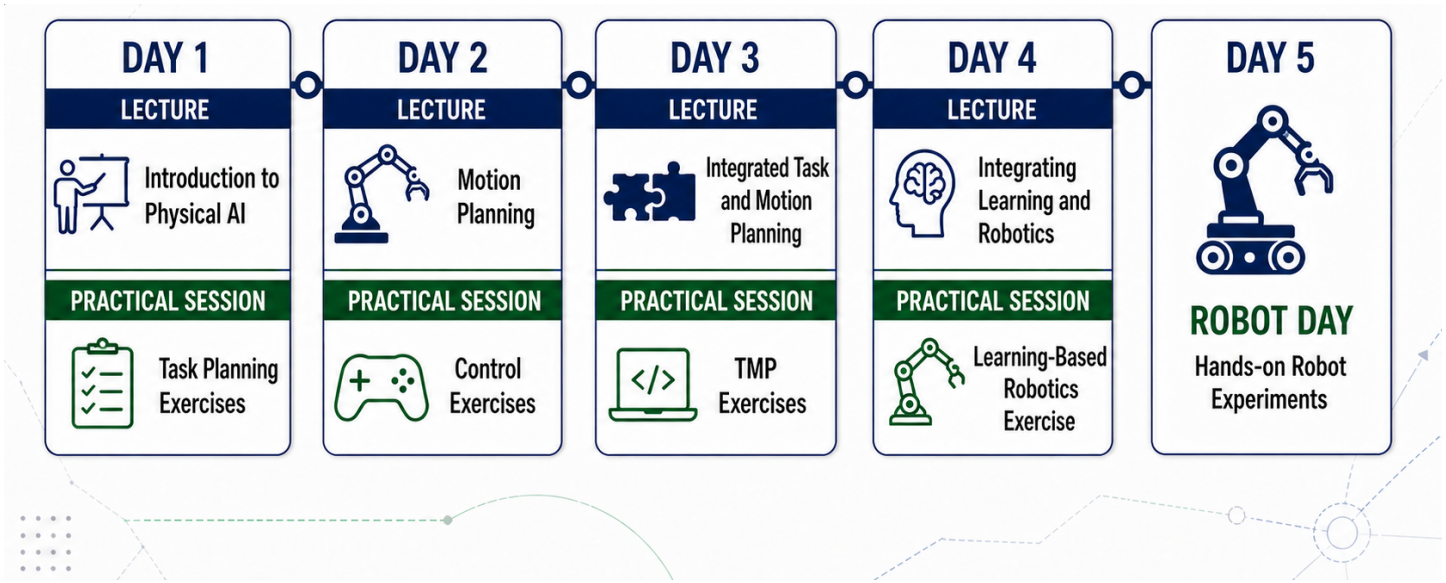
- **Reinforcement Learning:** The robot can learn grasp strategies through trial and error based on success rates.
- **Imitation Learning:** The robot can learn from demonstrations of humans picking apples.
- **Active Perception:** If uncertainty in pose estimation is high, the robot may reposition its camera to obtain a better view before acting.

Open Questions

- How to update the state representation?
- How to interface the three layers?
  - For example, which component maintains valid grasp configurations?
- How to account for uncertainty?



Agenda


DAY 1
LECTURE
Introduction to Physical AI
PRACTICAL SESSION
Task Planning Exercises
DAY 2
LECTURE
Motion Planning
PRACTICAL SESSION
Control Exercises
DAY 3
LECTURE
Integrated Task and Motion Planning
PRACTICAL SESSION
TMP Exercises
DAY 4
LECTURE
Integrating Learning and Robotics
PRACTICAL SESSION
Learning-Based Robotics Exercise
DAY 5
ROBOT DAY
Hands-on Robot Experiments

Clarifications

- Labs are intentionally more elaborate than we have time to complete.
- All materials will be available after the deadline.
- You will be welcome to come and work on the robots at the lab after the WS is over.

WS info

Website:
https://clair-lab-technion.github.io/physical-ai-workshop

Repo:
https://github.com/CLAIR-LAB-TECHNION/physical-ai-workshop